

Ch-6

Pressure is the force per unit area. SI unit for this is N/m^2 . This unit is known as Pascals (Pa).

Broad surfaces & narrow surfaces, & if have same mass & same amount of pressure is being exerted on them, then broader surface will apply less p but narrow surfaces apply more p.

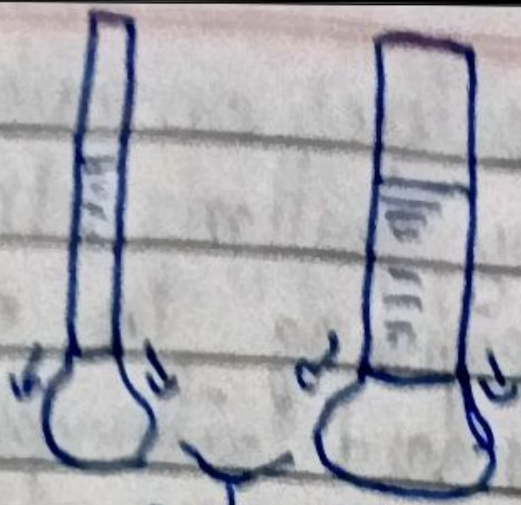
E.g.:

$$1 - \frac{100 \text{ N}}{25 \text{ m}^2} = 4 \text{ N/m}^2 = 4 \text{ Pa}$$

$$2 - \frac{100 \text{ N}}{10 \text{ m}^2} = 10 \text{ N/m}^2 = 10 \text{ Pa}$$

If there are 2 water columns of same height & they are filled with water to the same extent, then pressure will be the same under them.

Dia 1 =



Pressure
is same

Dia 2 =



Dia 2 is a proof that liquids exert pressure of every side.

The bottom of a dam is much broader than the top due to these 2 reasons:

- 1) Water exerts pressure on the bottom more than any point because the water in dam exerts horizontal p on side walls & vertical p on side.
- 2) It has to support the whole structure & make it strong.

The envelope of air surrounding the Earth is known as atmosphere. It consists of Nitrogen, Oxygen, CO_2 , etc.

Air also exerts pressure on every obj. from every side. This's why balloons inflate & become round by expanding in every direction. Air pressure on standard ~~atmospheric~~ sea level on a 1 cm^2 surface is $\sim 10\text{ N}$.

Units of Air pressure

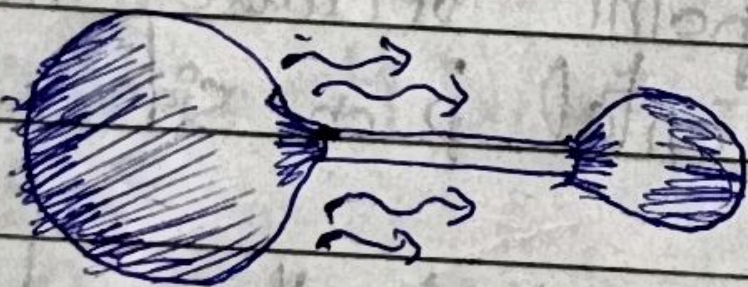
Milibar (mb)

Hectopascals (Hpa)

Pascals (Pa)

$$100\text{ Pa} = 1\text{ mb} / 1\text{ Hpa}$$

Dia 3:



In this setup, air moves from more inflated balloon to less inflated balloon. This proves that air moves from high pressure zone to low pressure zone.

DATE: _____
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Sea breeze & land breeze also occur due to this. In daytime, sun's heat heats up everything but land gets heated faster than water, so air above land, rises up & makes a LPZ. So, air above water rushes toward the land, this is known as sea breeze.

But, at night, this process reverses & wind blows from land to water & causes land breeze.

When high speed winds blow, they're accompanied by LPZ. So, during a storm, it is better to open windows & doors of the house because when high speed make a LPZ above the house, if that pressure difference is high & the roof is weak ~~there~~ then the roof might be blown away. So, to reduce the pressure diff. the window-n-door opening solution is a great idea.

[Nature's Phenomenon]

[Storm]

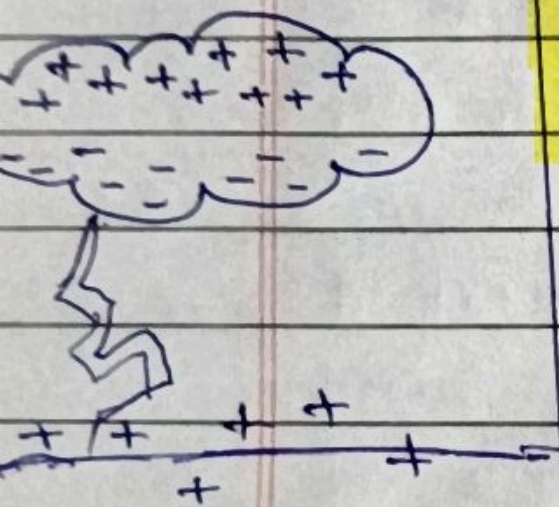
The high-speed wind accompanied by rain is called a storm.

[Lightning]

When strong winds blow upward & downward, facilitate rubbing b/w water drops or ice particles. This rubbing generates electrostatic force & causes dev. of static charges. Pos. charges move up & neg. charges move down. When this accumulation becomes very large, it breaks insulation of air & causes lightning. Lightning is the sudden flow of charge, producing a bright flash, which is called lightning.

[Thunderstorm]

The quick expansion of air due to lightning produces a sound, which is known as thunder. A storm accompanied by lightning & thunder is called a thunderstorm.



Lightning occurs within a cloud, between the clouds or b/w cloud & ground.

Lightning can be dangerous. To protect yourself from it:

- Stay away from tall obj
- Find a ~~to~~ low-lying open area
- Crouch but don't lie flat
- ~~Min~~ Minimize contact with ground
- Hide in a car or bus, if you can

Thunderstorms in India are known by various name in different regions. Like: Kalboishakhi (West Bengal) or Bordoisala in Assam. They help in mango growth in Kerala, Karnataka & Tamil Nadu. They also help in coffee growth in Karnataka.

A lightning conductors is a metallic rod placed on the highest point of a building, it provides a easy way for electric charges into the ground.

Cyclones are storms that form over warm oceans.

As water gets heated, warm & moist air rises upward & becomes raindrop.

When vapours become raindrop, that heat is released back into atmosphere.

This heat makes vapours to rise even higher.

This rising makes an LPZ, & causes surrounding ~~area's~~ area's ~~to~~ air to rushes toward that LPZ.

Earth's rotation causes that rushing air to spin. This ~~makes~~ a process is repeated again-n-again, which results in the creation of a area of even low pressure & high-speed winds revolving around it.

This system of clouds, winds & rain is ~~known~~ known as a cyclone. The centre of the cyclone is called the eye of cyclone & it is very calm. Amphan Cyclone reached the speed of 270 KMph.

Weather reports, warnings & alerts are issued by Indian Meteorological Department (IMD).